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Investigating the interaction of pretreatment and saccharification processes for bioethanol production from olive mill solid waste

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ABSTRACT

Statistical Design of Experiments was used to investigate the interaction between the pretreatment and saccharification processes for ethanol production from olive mill solid waste (OMSW). Sixty-four experimental runs were performed to test for four pretreatment factors: microwave (MW) temperature and duration, organic acid (OA) type and concentration, and two enzymatic factors: enzyme load and Tween 80 addition. Response variables were: loss of biomass and sugars during pretreatment, cellulose saccharification efficiency, formation of fermentation inhibitors, yeast growth, and ethanol production. Saccharification efficiencies were 17 %–94 %, and ethanol production was 6–15 g/L. Hydroxymethyl furfural (HMF) and furfural concentrations were less than 1.1 g/L throughout the runs. Within the investigated ranges, microwave temperature, OA type, and enzyme load had the most significant effects on ethanol production, with a strong interaction between OA type and temperature. Enzyme load had no interaction with other factors. Furthermore, although a maximum ethanol yield (13.4 g/L) required 180 °C, 0.6 mol/L maleic acid, and 22 FPU/g treated OMSW, 11.4 g/L ethanol can be achieved with only 0.4 mol/L maleic acid and 16.5 FPU/g enzyme load, lowering costs. A shorter MW treatment (4 min) and no Tween 80 addition can also be used, reducing costs further since these factors had no significant effect on ethanol yields. The key result was that pretreatment and saccharification processes can be optimized separately.

1. Introduction

Olive oil production is an important agro-industry. Nevertheless, olive oil production generates side streams, each ton of olives yielding only ~200 kg of oil, up to 0.6 tons of solid, and 1.5 m^3 of liquid side stream [1]. Management of these wastes is challenging economically, due to high treatment and processing costs, limited by-product markets, and seasonal production that increases storage and handling expenses, and environmentally, it can cause soil and water contamination due to high phenolic content and acidity [1,2]. Olive mill solid waste (OMSW) is lignocellulosic biomass, and attempts were made to treat OMSW by composting [3], as a substrate for edible fungi [4], as biochar (or biosorbent) feedstock [1,5], and as feedstock for biofuel production [1,2,6,7]. However, each treatment presents specific challenges. For the anaerobic digestion of OMSW, integrated strategies that combine efficient pretreatment, co-digestion, and sustainable processes are needed [8]. Despite these efforts, a viable solution has yet to be found.

Interest in ethanol production from lignocellulose biomass is increasing [9] optimization of this pretreatment is necessary for each type of biomass [10–12]. The pretreatment process is one of three main processes involved in bioethanol production from lignocellulosic biomass, i.e., pretreatment, hydrolysis (chemical or enzymatic), and fermentation. Various methods for Lignocellulosic biomass pretreatment have been studied, including physical, chemical, physicochemical, biological, and combination [13]. Different pretreatments were tested for the production of ethanol from OMSW, including acid-catalyzed

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organosolv [14]; dilute acid and Alkaline treatment [15]; basic pretreatment, and calcium carbonate addition [16]; dilute sulfuric acid with [17] and without a steam explosion [18], microwave irradiation [19]. However, OMSW is currently a side stream without a commercial, economic, or practical solution [20–22]. Therefore, there is a need to optimize the pretreatment process in order to increase the ethanol production. Microwave-assisted physicochemical pretreatment for converting lignocellulosic biomass into biofuels has attracted significant attention [23–25]. While sulfuric acid is commonly used in chemical pretreatment, one drawback is that it generates toxic inhibitors to microorganisms in the fermentation process [26]. Many studies have investigated the use of organic acids in pretreatment, like maleic acid or formic acid, to overcome this lack. It has been found that microwave treatment-assisted formic acid is effective [1], and using maleic acid generates fewer inhibitors than sulfuric acid. It increases the production of glucose and xylose monomers. Organic acid pretreatment is considered the most suitable method of commercial treatment and is significant for achieving cleaner production and energy utilization. Formic acid pretreatment effectively separates lignin due to its similar solubility parameters. The acid can be easily recovered through simple distillation because of its low boiling point [27]. In recent years, there has been extensive research on the effects and mechanisms of formic acid pretreatment. It has been found that formic acid exhibits high delignification efficiency and selectivity. For example, Corn cob was hydrolyzed at 80 °C for 3 h with formic acid pretreatment. The separation yield of lignin was 87.1 ± 3.4 % [27].

Maleic acid is a non-toxic acid [28], and has a positive effect on separating lignin [29], and 88.0 % efficiency of lignin separation [30]. Previous works showed that the ethanol fermentation yield was effectively improved [31], and the content of residual condensed lignin decreases, which facilitates further catalytic oxidation [32].

Microwave irradiation is well-established in food processing and green chemistry processes [33]. It has many potential merits for lignocellulosic biomass pretreatments, among them being contact-free, heating evenly, having a short reaction time, limited side reactions, and simple parametric variations [34]. Nevertheless, such use is still very limited, and most research was carried out on MW use for pyrolysis [35]. Microwave-based biomass pretreatment was demonstrated for different feedstocks, including hardwoods and softwoods, sugarcane bagasse, rice straw, wheat straw, and switchgrass biomass [36]. Microwaves promoted an increase in the removal of non-cellulosic compounds in the eucalyptus waste [37]. When compared with conventional approaches, the pretreatment approaches using microwave would usually possess a faster reaction rate with less energy [38,39].

It is well-known that reaction conditions, such as temperature, time, catalyst type and amount, biomass loading and type, and MW power, significantly impact yields and are critical parameters of the reaction products [40–43]. To our knowledge, optimizing the use of microwave-assisted pretreatment combined with organic acids (maleic and formic acid), followed by enzymatic saccharification and fermentation, is relatively underexplored, particularly for OMSW. Where the study of Abu Tayeh et al., 2020, focused on the effects of OMSW different pretreatments (microwave or autoclave), different additives (water, formic, or sulfuric acid), and utilization of different saccharification enzymes (Cellic_ CTec2 or Accellerase_ 1500) [1], and the work of El-Bashiti et al., 2017, focused on the production of ethanol from olive solid waste by different *Saccharomyces cerevisiae* isolates after pretreatment by physical methods and subjected to hydrolysis by microwave-assisted 5 % HCl procedures [44]. On the other hand, the optimization of both the pretreatment and the enzymatic saccharification processes and the interaction between these two stages for OMSW biomass have not been investigated previously.

Our research hypothesis is that the interaction between different pretreatment and saccharification processes has no significant effect on optimizing ethanol production from the OMSW. Therefore, our study objective was to investigate the interaction between the pretreatment and saccharification processes for ethanol production from OMSW using Design of Experiments (DoE).

2. Materials and methods

Here, we have used the DoE approach to investigate different parameters of MW and organic acid (OA) pretreatment, including temperature, treatment duration, and various OAs. The OA types included formi [1,45], and maleic acid, also demonstrated for use at different concentrations [46,47]. We have also tested the effect of enzyme load (enzymes are a major cost factor) and the addition of Tween 80 surfactants to the enzymatic saccharification stage, as the latter was previously reported to improve saccharification, but never tested on OMSW, to our knowledge [48,49].

2.1. Experimental design and setup

The Design of Experiments (DoE) approach was used in the design and analysis of the experiment, using an optimal design constructed by the Custom Design tool in the statistical software JMP 15 (SAS Institute). The design was a split-plot design with six factors (four factors for the pretreatment process and two factors for the enzymatic saccharification process), run over four days of pretreatment and two days of the saccharification process (Figs. 1 and 2). The experimental data for each response factor were used to fit statistical models. The starting point was an initial model simplified by backward elimination: removing terms with no significant contribution ($p > 0.05$) sequentially and then refitting the model with the remaining terms. All factors in our JMP design were analyzed using coded variables, meaning that each factor, regardless of its raw units, was scaled to an identical interval (−1 to +1). Therefore, the estimated effects and significance levels reflect the influence of each factor over an equivalent relative range, not the absolute physical range. Further discussion of the details of the experimental design, in particular the split-plot structure, the inclusion of control runs, and the reasons why the number of runs is not identical for all experiments, is included in section (S1).

2.2. Olive mill solid waste (OMSW)

The OMSW was collected during the fall of 2018 from a 3-phase olive oil production process located in Reineh village in the Galilee region of Israel. The OMSW was dried under shade at ambient temperature for a week, milled using a coffee grinder, and screened through an 18-mesh screen, and the <1 mm particles were stored in tightly sealed plastic bags at room temperature until use. The OMSW powder composition was analyzed using the standard analytical procedure described in NREL [50]. Cellulose, hemicellulose, lignin, extractives, ash, and acetyl content of the OMSW biomass are presented in Table 1.

2.3. Pretreatment process

The microwave (START D, Milestone S.r.l) has a multi-vessel rotating reactor system equipped with a cartridge with 12 vessels, each with a capacity of 100 mL. Eight of these vessels were used for each sub-batch, meaning that these groups of eight runs had the same temperature and time level (Fig. 2). A built-in ventilation system enabled quick cooling. A pyramid-shaped rotating diffuser in the reactor cavity ensured the even distribution of the microwave irradiation. Five-gram solids in 50 mL of liquid were used in all runs (resulting in 1:10 w/v). Four pretreatment factors, pretreatment temperature, pretreatment time, OA type, and OA concentration, were varied, as shown below in Table 2 and Fig. 2. Maleic and formic acids were diluted from the concentrate (Sigma-Merc, USA) into deionized water.

After the pretreatment process, every 64 runs were vacuum filtered through the Whatman 1 paper filter, and the solid fraction was dried at room temperature to 10 % moisture and used for enzymatic

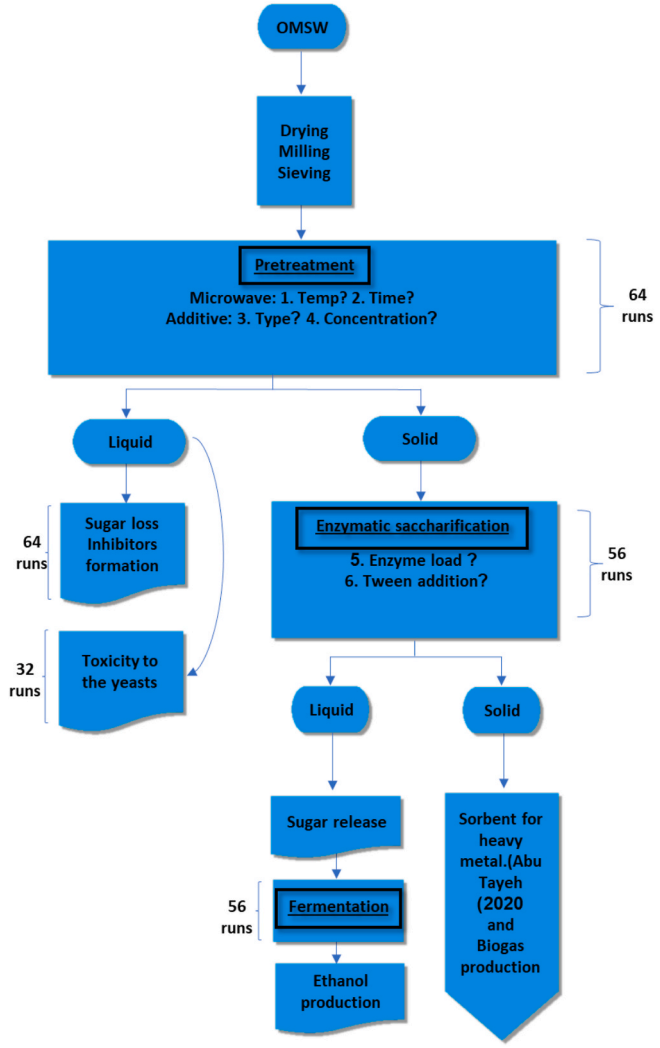


Fig. 1. The study flow chart and experimental logic. There are three main stages for ethanol production from OMSW; the first is the pretreatment using a microwave combined with organic acid. Liquid produced from this stage is analyzed for sugar content, inhibitor formation, and potential toxicity to yeasts. Second is enzymatic hydrolysis for the solid phase of the previous stage, with enzyme load and Tween addition. The solid from this stage is used as a sorbent for heavy metals and biogas production. Moreover, the third is the fermentation process of the liquid phase of the previous stage of ethanol production.

saccharification. The filtrate was analyzed for sugars, furfural, and HMF by HPLC-RI using HPLC (Thermo Scientific Finnigan Surveyor), equipped with a Rezex™ ROA-Organic Acid H^+ column (150 × 7.8 mm; Phenomenex) (as in Ref. [1]). Briefly, the Column temperature was held at 68 °C, with isocratic 0.005 N sulfuric acid in water at a flow rate of 0.6 mL/min. Compounds were identified by comparing retention time to standard Sugars using a refractive index detector (Shimadzu) and fermentation inhibitors using a diode array detector (Thermo) set to 280 nm.

Our study focuses exclusively on the enzymatic production of glucose for ethanol production; therefore, the pretreatment filtrate was not utilized for ethanol production and was considered a loss of sugar.

Sugar loss into the pretreatment liquid and sugar release by enzymatic saccharification were calculated as follows [51]:

$$Gt = Gp + 1.43 \times Hp \quad (1)$$

$$Pt = XPp + 1.56 \times Fp \quad (2)$$

$$Gy = \frac{Gh \times 100}{Gf \times B} \quad (3)$$

Where *Gt* is the total glucose (g/L); *Gp* is glucose concentration in the liquid after the pretreatment process (g/L); *Hp* is HMF concentration in the liquid after the pretreatment process(g/L); 1 g HMF is formed from 1.43 g of glucose; *Pt* is a total pentose(g/L); *XPp* is xylose/arabinose concentration in the liquid after the pretreatment process(g/L); *Fp* is furfural concentration in the liquid after the pretreatment process(g/L); 1 g of furfural is formed from 1.56 g of xylose or arabinose; *Gy* is glucose yield(%) after the enzymatic saccharification process; *Gh* is glucose concentration after the enzymatic saccharification process (g/L); *Gf* is Glucose fraction in treated OMSW; *B* is dry biomass concentration (g/L).

These value ranges of experiment parameters were determined based on previous works.

2.4. Toxicity to yeast of the pretreatment liquid fraction

The filtrate of the pretreatment liquid fraction was tested for its effect on yeast growth using the agar well diffusion assay [52]. Briefly, Ethanol Red® yeast (Fermentis) was grown in YPD broth (10 g/L yeast extract, 20 g/L peptone, 20 g/L glucose) until mid-log, smeared on YPD agar (same as the YPD broth with the addition of 20 g/L agar), and a hole was bored in the agar using an ethanol sterilized and flamed 5 mm cork borer. The hole was filled with pretreatment filtrate, plates were placed in an anaerobic box, an anaerobic pack (MGC, Japan) was dropped in, and the box was incubated at 35 °C for 48 h. Inhibition by the filtrate for each run was measured as the average diameter of the no-growth zone around the well for three wells prepared from a single sample.

2.5. Enzymatic saccharification

Enzymatic saccharification was done using the Cellic CTec2 enzyme (Novozyme). An enzyme cocktail contains cellulases, beta-glucosidases, and hemicellulose that are well known for efficiently hydrolyze lignocellulosic biomass into fermentable sugars. Three grams of each pretreatment solid fraction was placed in a separate 50 mL Erlenmeyer flask, and 18.75 mL of 50 mmol/L Na-citrate buffer at pH 5.2 was added (resulting in a w/v ratio of 16 %). The flasks were incubated for 24 h at 50 °C, shaking at 170 rpm. Enzyme load and Tween 80 were varied, as shown in Table 2 and Fig. 2B. Saccharification of 56 of the 64 pretreated samples was carried out, as eight "spare" samples were reserved, one from each sub-batch. After 24 h, the hydrolysate was cleared of solids by centrifugation (4000 g, 10 min, room temperature), and the liquid fraction was frozen until used for fermentation. HPLC-RI also analyzed the liquid fraction for sugar content [1], listed as Glucose Hydrolysate (g/L) in Table 2.

2.6. Fermentation

The liquid phase of the saccharification process was used as feedstock for fermentation. Ethanol Red® (Fermentis) yeast was cultivated on YPD agar for 24h at 35 °C. A single colony was transferred to YPD broth and incubated shaking at 110 rpm, at 35 °C overnight. Two mL of inoculum were centrifuged (2 min, 6000g, at room temperature), the liquid decanted, and the pellet was re-suspended in the remaining liquid (~100 μL). The cell suspension was added to 15 mL hydrolysate in a 50 mL Erlenmeyer, the gas phase was flushed briefly with nitrogen, and the Erlenmeyer was closed with rubber septa and incubated at 35 °C, 120 rpm. After 24h, samples were taken, cells were removed by centrifugation (8000 g for 5 min), and the liquid fraction was analyzed by HPLC for ethanol and sugars. The ethanol efficiency (EE) was calculated as follows:

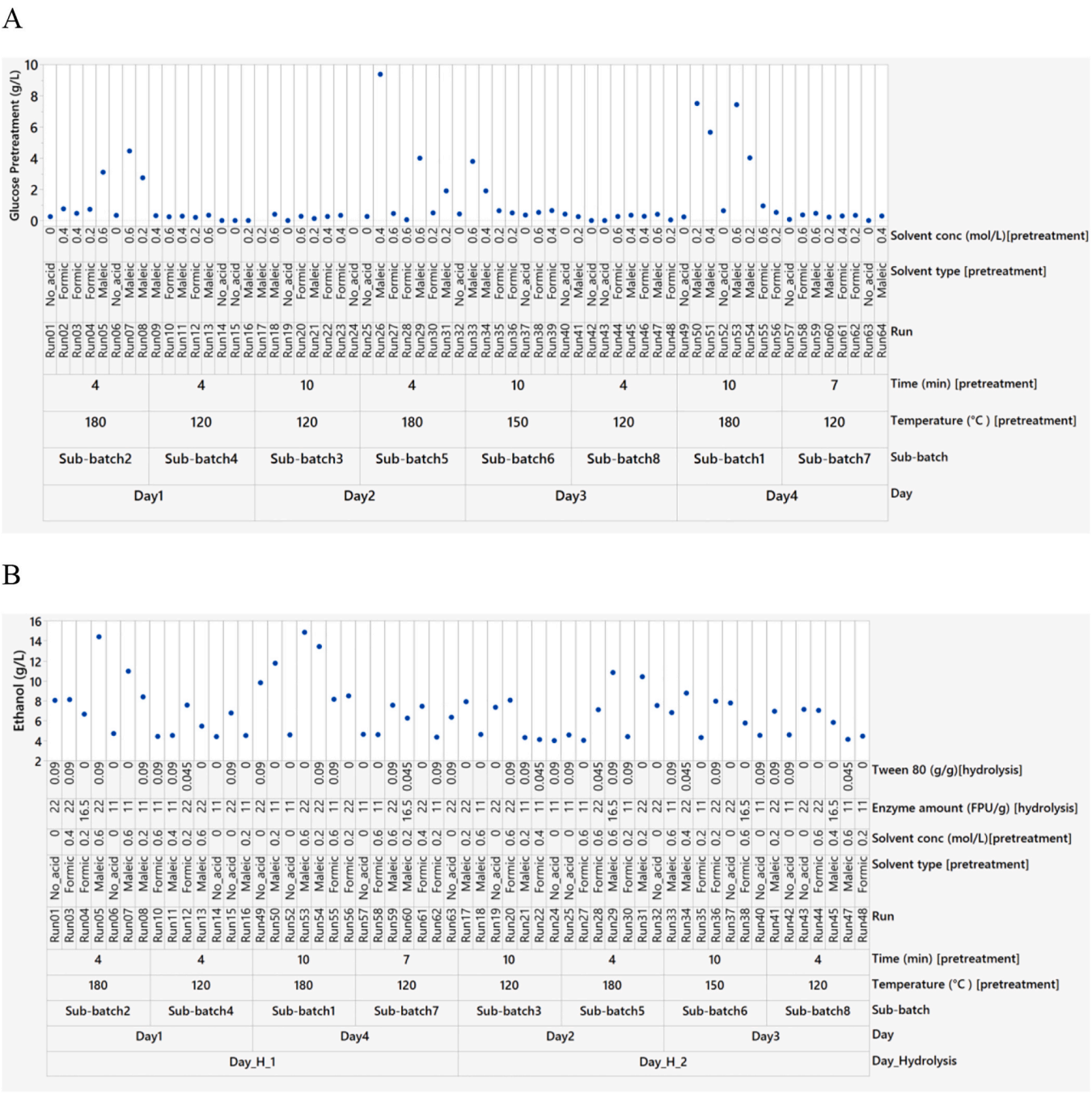


Fig. 2. Graphical display of experimental design and raw data for (**A**) Glucose loss after pretreatment for each combination of temperature, time, OA type, and OA concentration and (**B**) ethanol production for each combination of temperature, time, OA type, OA concentration, enzyme load and Tween 80, both showing sub-batches and days.

Table 1
Analysis results of OMSW compositions, mean values of three replicates.

Parameter	(%)[a] ± STDEV
Cellulose	27.0 ± 2.0 %
Hemicellulose	17.3 ± 0.5 %
Lignin	30.0 ± 5.0 %
Extractives	19.0 ± 1.0 %
Ash	4.3 ± 1.3 %
Acetyl content	2.1 ± 0.4 %

[a] Based on DW.

$$EE\ (\%) = [Ethanol\ (g/L)] \times 100 / (Glucose\ fraction\ in\ treated\ OMSW \times biomass \times 0.51) \quad (4)$$

Where [Ethanol (g/L)] is ethanol concentration after the fermentation process; biomass is dry biomass concentration (g/L); 0.51 is the conversion factor for glucose to ethanol.

It is important to mention that our objectives did not include fermenting the hydrolysate after the pretreatment process.

Table 2
Structure of the experiment: number of runs, factors, and responses for each stage of the process.

Stage of process	Number of runs	Number of factors (input parameters)	Responses (outcomes)	Measure	Results in section
After pretreatment	64 (62 with results; 47 with acid, 15 with no acid)	Four factors[a]	Glucose Pretreatment_Liquid (g/L)	Loss of sugars	3.1
			Xylose Pretreatment_Liquid (g/L)	Loss of sugars	3.1
			Arabinose Pretreatment_Liquid (g/L)	Loss of sugars	3.1
			Furfural Pretreatment_Liquid (g/L)	Formation of inhibitors	3.1
			HMF Pretreatment_Liquid (g/L)	Formation of inhibitors	3.1
			Loss of biomass	Loss of biomass	3.2
After pretreatment	32	Four factors[a]	Yeast inhibition (mm)	Toxicity to yeast	3.3
After enzymatic saccharification	56 (16 with no acid)	All six factors[b]	Glucose Hydrolysate (g/L)	Cellulose saccharification	3.4
After fermentation	56 (16 with no acid)	All six factors[b]	Ethanol (g/L)	Production of ethanol	3.4

[a] Temperature at 120, 150, and 180°C; time at 4, 7, and 10 min; OA type at No_acid, formic, maleic; OA concentration at 0.2, 0.4, 0.6 mol/L for OA type = formic or maleic.
[b] The four factors listed above and enzyme load at 11, 16.5, 22 FPU/g treated OMSW and Tween 80 at 0, 0.045, and 0.09 g/g treated OMSW.

3. Results and discussion

3.1. Effect of pretreatment conditions on the loss of sugars and fermentation inhibitors formation

For all five responses, glucose, xylose, arabinose loss, furfural, and HMF, models were fitted separately to the main part of the experiment with OA (with OA concentration as a factor) and to the control runs with no OA. Fig. 1S includes a graphical display of the raw data showing correlations between responses.

The control runs with no OA had relatively low glucose, xylose, and arabinose loss (the highest observed level was 1.3 g/L, equivalent to 7.7 % of the hemicellulose fraction) and no furfural or HMF. Here, glucose, xylose, and arabinose loss were all found to have a clear positive dependence on temperature.

For the main part of the experiment with added OA (47 runs), a fitted model for each of the five responses was obtained by sequentially removing insignificant effects ($p > 0.05$) from the full initial model (Table 1S). Common to all fitted models for the main part of the experiment is the importance of temperature and OA type. The fitted model with OA for each of the five responses is written as two separate prediction equations for MA and FA (Table 3), and the response surface plots for all five responses are shown in Fig. 3. The version of the fitted model with centered polynomials is given in Table 3 for ease of interpretation. The random sub-batch effect is included due to the split-plot design, the choice of which was dictated by the structure of the experimental process, as discussed in S1. The full models included all main effects, interactions, and quadratic effects, as reported in Table 1S. A full report of the analysis for each model, including details of which effects were significant and of the exclusion of three measurements suspected to be outliers, is included in Table 1S and section S2. The experimental design enables quadratic effects to be included in the full models, as in a classical response surface design, but only one quadratic effect for one response is significant and hence included in the fitted models.

As presented in Fig. 3, for glucose loss there is a strong interaction between Temperature and OA type: glucose loss was strongly dependent on temperature in the presence of MA (Table 3, Fig. 3B; at 10 min, the predicted level of glucose loss is 0.7 g/L at 120 °C and 5.7 g/L, 25 % of the cellulose fraction, at 180 °C), and less so with FA (Table 3, Fig. 3B; only reaching 1.4 g/L, 4 % of the cellulose fraction), both much lower than previously shown with sulfuric acid (20 %; [1]). Xylose loss (Fig. 3B) was similarly dependent on temperature (MA ≫ FA; higher temperature resulted in higher xylose loss). Arabinose loss showed a somewhat different pattern (MA ≳ FA; higher temperature resulting in higher arabinose loss) (Fig. 3C). When the OMSW was extracted with water (100 °C, 30 min) and pretreated at 170 °C, using 2 % H_2SO_4 (w/v) and a 20 % substrate concentration in a 1 L Parr reactor the composition of the glucose, xylose, and arabinose in the filtrate were 4.63 g/L, 23.69 g/L, and 3.42 g/L, respectively [53]. The response surface results, Fig. 3, indicate that the interactions between factors were weak, with linear effects of temperature and time dominating the fitted model. This suggests that each factor contributed additively to glucose release. Such linearity simplifies optimization within the studied range, though broader factor domains may reveal more complex interactions.

Thermal pretreatment processes often release byproducts such as furfural, hydroxyethyl furfural (HMF), and a variety of organic acids, implying the degradation of sugars [53,54]. The results presented in Fig. 3D and E demonstrate that Furfural production was highly

Table 3
Fitted models displaying the effect of pretreatment factors (temperature, time, OA type, and concentration) on loss of glucose, loss of xylose, loss of arabinose in the liquid phase for the treated OMSW, formation of furfural, and HMF.

OA type	Fitted models
Maleic acid	Glucose loss (g/L) = −9.6090 + 0.0707 × Temperature + 0.2235 × Time + 0.0044 × (Temperature – 146.0870) × (Time – 6.6090) Xylose loss (g/L) = −22.0512 + 0.157 × Temperature + 0.3750 × Time +2.6870 × OA conc. + 0.0066 × (Temperature – 146.8100) × (Time – 6.5500) Arabinose loss (g/L) = −3.5270 + 0.0350 × Temperature + 0.1075 × Time + 0.0021 × (Temperature – 146.8100) × (Time – 6.5500) Furfural Pretreatment_Liquid (g/L) = −1.5310 + 0.0120 × Temperature + 0.2980 × OA conc + 0.0110 × (Temperature – 146.8100) × (OA conc – 0.4000) HMF Pretreatment_Liquid (g/L) = −0.2860 + 0.0020 × Temperature + 0.0060 × Time + (Temperature – 146.0870) × (Temperature – 146.0870) × 0.0001 + 0.0002 × (Temperature – 146.0900) × (Time – 6.4800)
Formic acid	Glucose loss (g/L) = −0.6612 + 0.00719 × Temperature + 0.0025 × Time + 0.0044 × (Temperature – 146.0900) × (Time – 6.6100) Xylose loss (g/L) = −3.0820 + 0.0170 × Temperature + 0.0658 × Time + 2.6870 × OA conc. + 0.0066 × (Temperature – 146.8100) × (Time – 6.5500) Arabinose loss (g/L) = −4.5600 + 0.0350 × Temperature +0.1075 × Time + 0.0021 × (Temperature – 146.8100) × (Time −6.5500) Furfural Pretreatment_Liquid (g/L) = −0.5150 + 0.0030 × Temperature + 0.2980 × OA conc + 0.011 × (Temperature – 146.8100) × (OA conc – 0.4000) HMF Pretreatment_Liquid (g/L) = −0.1310 + 0.0005 × Temperature + 0.0060 × Time + (Temperature – 146.0900) × (Temperature – 146.0900) × 0.0001 + 0.0002 × (Temperature – 146.0900) × (Time – 6.4800)

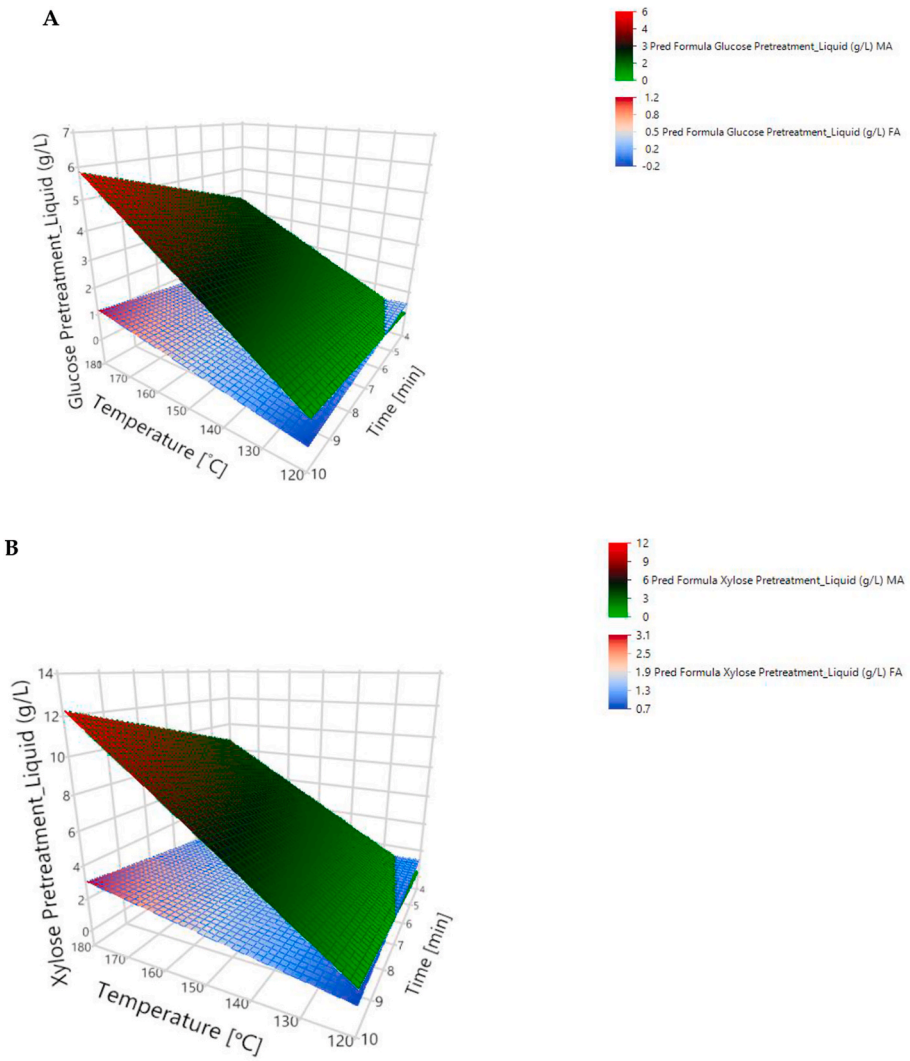


Fig. 3. (A–E) Response surface plots displaying the effect of pretreatment factors (temperature, time, OA type, and concentration) on (A) loss of glucose, (B) loss of xylose, (C) loss of arabinose to the pretreatment liquid, formation of (D) furfural, and (E) HMF. The top surface is for MA, the bottom surface is for FA. Where no acid concentration is designated, it was set at 0.6 M. Note that all models include only statistically significant effects.

dependent on temperature and OA type (MA resulting in more furfural than FA), but to a lesser extent on OA concentration, and not at all on pretreatment time (Fig. 3D). Production of HMF (Fig. 3E) increased significantly with temperature and showed some dependence on pretreatment time (Fig. 3E) and on OA type, but not on OA concentration. It is noteworthy that the highest concentration of furfural observed was only 1.0 g/L and that of HMF 0.19 g/L, both well below the 30 m mol/L concentration known to be toxic to yeasts ([55]; 2.9 g/L and 3.78 g/L for furfural and HMF, respectively). It is also considerably lower than the previously published levels for maleic acid reported by Ref. [56](1.8 g/l furfural for corn stover) and by Ref. [55](3 g/L furfural), and similar to Ref. [1], i.e., 1.9 g/L furfural and 0.16 g/L HMF, and 1.488 g/L furfural and 0.1887 g/L HMF, where the OMSW was subjected to thermal pretreatment in a reactor for 60 min at 170 °C and 0.85 MPa pressure by direct vapor injection with a ratio of 1.1 L_{water}/kg_{OMSW} [57], and 1.95 g/L furfural and 0.15 g/L HMF were reported in another study where the OMSW was extracted with water (100 °C, 30 min) and pretreated at 170 °C, using 2 % H_2SO_4 (w/v) and a 20 % substrate concentration in a 1 L Parr reactor [58]. The production of toxic compounds is not due to the hydrolysis process but due to thermal pretreatment; previous work quantified the diversity of fermentation inhibitors from the thermal treatment of OMSW [1] Moreover, demonstrated that the levels of such are low in the hydrolysate. So, although fermentation inhibitors might be a problem in some cases, they were not here.

3.2. *Effect of pretreatment conditions on loss of biomass*

For the response loss of biomass, models were fitted separately to the main part of the experiment with OA (with OA concentration as a factor) and to the control runs with no OA. The control runs with no OA had a low biomass loss (the highest observed level was 26.8 %), similar to FA and lower than MA. Here, biomass loss was found to have a clear positive dependence on temperature. For the main part of the experiment with OA, a fitted model for loss of biomass response was obtained by sequentially removing insignificant effects ($p > 0.05$; Table 1S) from the full initial model (Table 1S). Again, temperature and OA type were the most important factors. The fitted model with OA for the response loss of biomass is written as two separate prediction equations for MA and FA (Table 4), and the response surface plots for loss of biomass as two separate plots for MA and FA (Fig. 4B). The full model included all main effects, interactions and quadratic effects, as reported in Table 1S.

Loss of biomass after pretreatment was strongly dependent on temperature in the presence of MA (Table 4, Fig. 4; at 0.6 mol/L, the predicted level of Loss of biomass is 21.7 % at 120 °C and 41.6 % at 180 °C),

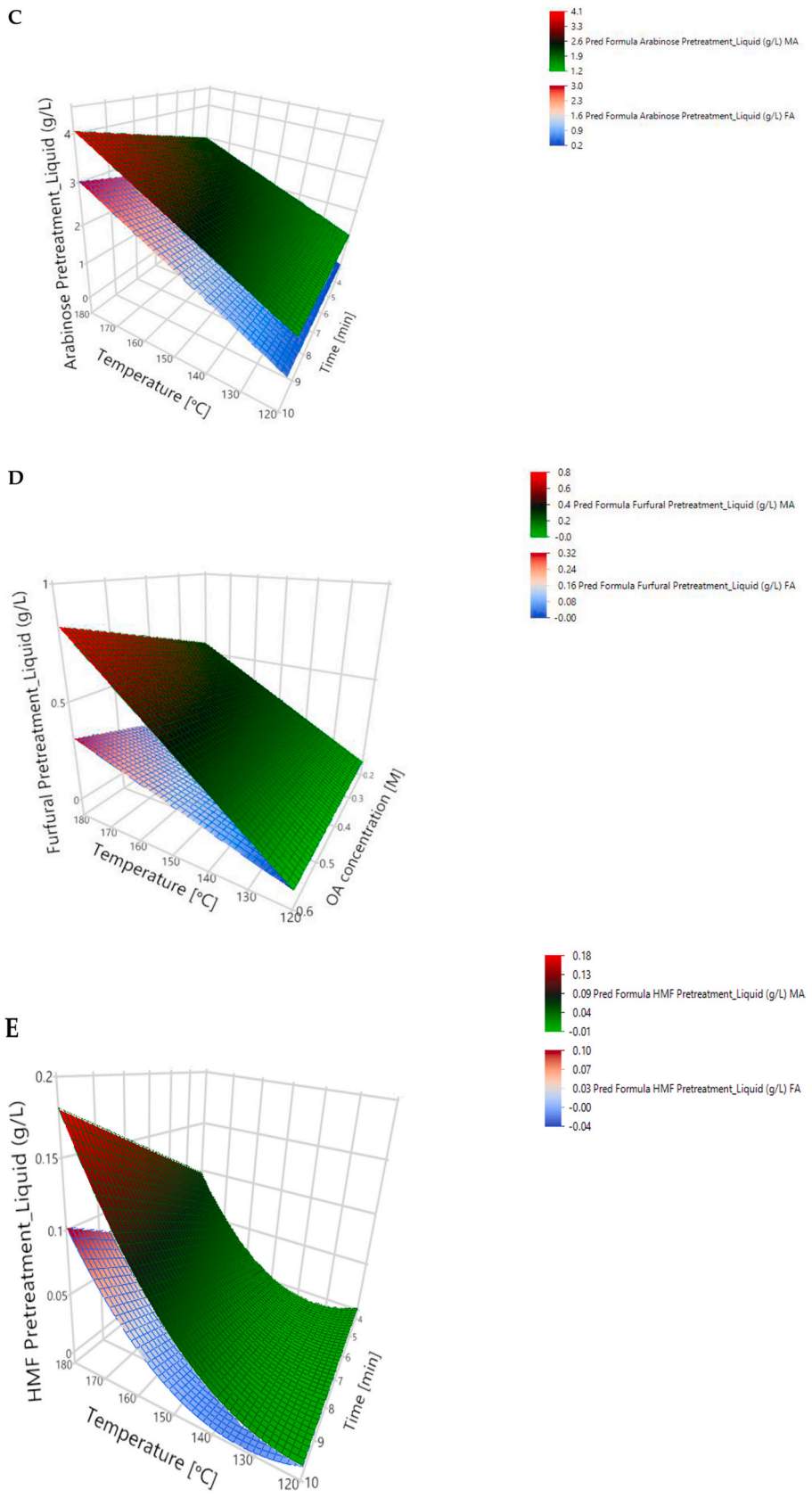


Fig. 3. (*continued*).

and less so with FA Table 4, Fig. 4; reaching 29.1 %). This result agrees with that reported on Water Hyacinth biomass treated with maleic acid, which showed a 15–50 % loss of biomass [59].

The results indicated that Maleic acid had the most significant impact on the loss of biomass, while formic acid had a mild effect. This could be due to their strength, i.e., maleic acid has a lower pKa than

Table 4
Fitted models displaying the effect of pretreatment factors (Temperature, OA type, and concentration) on biomass loss after the pretreatment process.

OA type	Fitted models
Maleic acid	Loss of biomass (%) = −20.273 + 0.291 × Temperature + 0.518 × Time + 4.962 × OA conc + 0.201 × (Temperature – 146.250) × (OA conc – 0.400)
Formic acid	Loss of biomass (%) = −10.568 + 0.167 × Temperature + 0.518 × Time + 4.962 × OA conc + 0.201 × (Temperature – 146.250) × (OA conc – 0.400)

formic acid. The pKa of organic acids plays a significant role in lignocellulosic biomass pretreatment, influencing the efficiency of hemicellulose and lignin solubilization. Organic acids with lower pKa values tend to be more effective in breaking down the complex structure of lignocellulosic biomass, leading to better yields of fermentable sugars. Maleic acid's first pKa (~1.9) is low enough to generate substantial proton activity for hydrolysis, but its second dissociation ($pKa_2 \approx 6.2$) and overall molecular structure confer enhanced selectivity and reduced sugar degradation [60]. Formic acid, with a higher pKa (~3.7), making it a relatively weak acid, less selective, and more prone to side reactions like cellulose formylation. A similar finding was also reported on the wheat straw samples that were pretreated with maleic, oxalic, and succinic acid at different temperatures (170, 190, and 210 °C), acid concentrations (1, 3, and 5 %), and times (10, 20, and 30 min) [61]. The xylan content of untreated wheat straw was 15.93 %, whereas the xylan content of maleic, oxalic, and succinic acid-treated wheat straw was in the range of 0.88–10.29 %, 0.07–7.13 %, and 6.60–20.33 %, respectively. In general, it could be concluded that the organic acid pretreatment removes xylan into the liquid phase.

Time had a negligible effect on the loss of biomass, with a slight increase in predicted loss of biomass (3.11 %) when time is increased from 4 to 10 min (coeff 0.518 × (10-4) = 3.11 %). Time had no significant interaction with any of the other factors, so increasing it results in an identical estimated increase for all levels of the other factors. The response surfaces in Fig. 4 have time fixed at a high level (10 min); lowering the time will give surfaces with identical shapes but slightly lower predicted responses.

3.3. Effect of pretreatment conditions on toxicity of the pretreatment liquid to yeasts

The yeast inhibition test was conducted on 32 of the 64 liquid fractions after pretreatment (Fig. 1), selected to enable the same full initial models to be fitted. Yeast inhibition was observed in 10 of the 12 runs with FA, but for none of the MA or no acid runs. The analysis of the 12 runs with FA showed a quadratic dependence of yeast inhibition on OA concentration (Fig. 5) and no dependence on temperature or time. Surprisingly, no yeast inhibition was observed for the runs with MA despite furfural and HMF being found to be higher for MA than for FA since both compounds are expected to inhibit yeast growth (Fig. 3). A possible explanation for the inhibition for FA despite lower levels of furfural and HMF, is the presence of FA residues (11–27 g/L), a primary inhibitory compound, in the liquid fraction. Detailed results are included in Supplementary Data 1S and Fig. 2S. The study by Zeng et al. (2022) found that cultivating yeast cells with 1.8 g/L of formic acid inhibited their ability to utilize sugar during the initial stages of fermentation, resulting in a slowed rate of alcohol production. The researchers demonstrated that Formic acid can induce oxidative stress, hinder protein synthesis, and cause DNA damage in S. cerevisiae, all of which may contribute to inhibited cell growth. Furthermore, the formic acid treatment led to elevated levels of certain aromatic amino acids, a decrease in nucleotide biosynthesis, and reduced energy consumption [62].

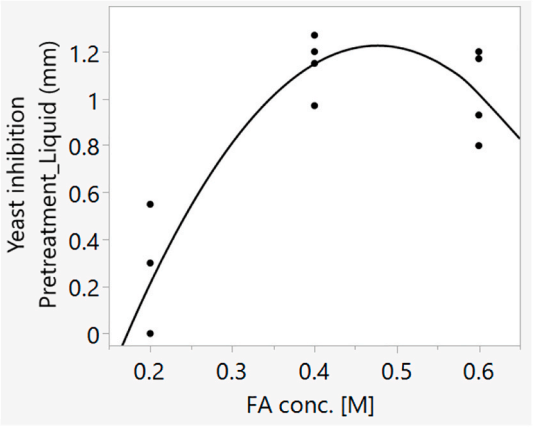


Fig. 5. Effect of FA concentration in the pretreatment liquid on yeast inhibition.

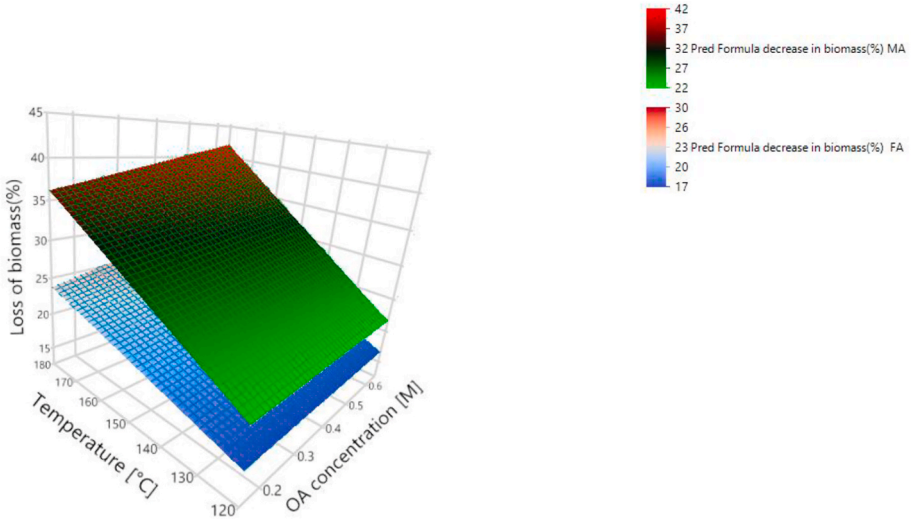


Fig. 4. Response surface plot displaying the effect of pretreatment factors (Temperature, OA type, and concentration) on the loss of biomass after the pretreatment process. For this display, the pretreatment time is fixed at 10 min.

3.4. The effect of pretreatment and saccharification conditions on cellulose saccharification and ethanol production

Enzyme load and tween amount factors varied in the enzymatic saccharification stage, following the pretreatment stage, where four factors varied, so the models fitted to glucose hydrolysate and ethanol responses included a total of six factors. As glucose hydrolysate (product of cellulose saccharification) and ethanol production were very highly correlated for the runs with OA (r = 0.99, Fig. 6A), their fitted models have a practically identical dependence on the factors. For this reason, the analyses for these two responses are displayed and discussed together. In addition, this very high correlation indicates that the saccharification process could be optimized in future experimentation without completing fermentation. However, this should be tested for

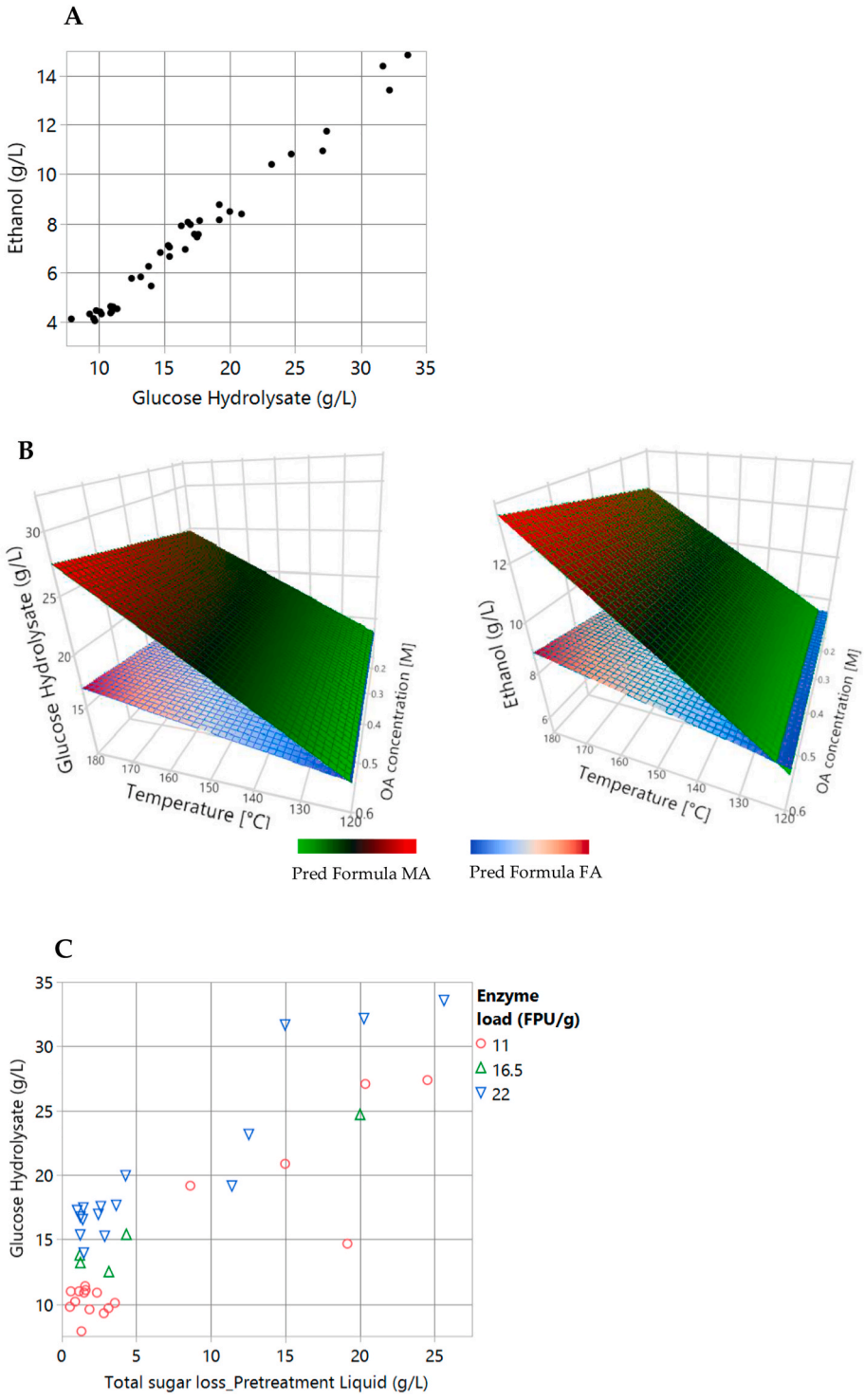


Fig. 6. (A) Ethanol and glucose hydrolysate correlation, (B) response surface plots displaying the effect of pretreatment parameters on glucose released after enzymatic saccharification (left plot) and on ethanol production (right plot) of OMSW treated with maleic acid (top surface), formic acid (bottom surface). Enzyme load set at 22 FPU/g OMSW. (C) Glucose hydrolysate vs total sugar loss in the pretreatment (runs with OA), by enzyme load, total sugar loss calculated according to Eqs. (1) and (2).

each biomass pretreatment separately and could be affected by the separation of solids from liquids after the pretreatment stage. As in the analyses of the previous stages, models were fitted separately to the main part of the experiment with OA (with OA concentration as a factor) and to the control runs with no OA.

As in the analyses of the pretreatment stage, the no_acid control runs had relatively low levels for both responses. Both glucose hydrolysate and ethanol responses depended on pretreatment temperature and enzyme load. With no OA, the highest levels of glucose hydrolysate and ethanol were observed with the temperature at a high level (180 °C) and high enzyme load (22 FPU/g), with averages of the three no_acid runs being 18.9 g/L glucose and 8.45 g/L ethanol. For the main part of the experiment with OA, a fitted model for each response was obtained by sequentially removing insignificant effects ($p > 0.05$) from the full initial model, which included all main effects, interactions, and quadratic effects for all six factors. The two fitted models found for glucose hydrolysate and ethanol are written as four prediction equations in Table 5, with each fitted model written separately for MA and FA. Response surface plots for the two responses are shown in Fig. 6B for enzyme load fixed at a high level (22 FPU/g), separately for MA and FA. A discussion of the meaning of the prediction equations and the response surface plots follows. Again, details are included in Table 1S and section S1.

With OA, glucose hydrolysate and ethanol were strongly dependent on temperature, OA type, concentration, and enzyme load, with all effects highly significant ($p < 0.01$). The pretreatment duration had an insignificant effect on the glucose hydrolysate in our study, contradictory (or opposite results obtained in another study that dealt with olive stone as substrate, showing that increasing in pretreatment time increases the glucose hydrolysate from the saccharification process [63]. Enzyme load had no significant interaction with any other factors, so increasing it results in an identical estimated increase for all levels of the other factors. A previous study showed that increasing cellulase load increased the D-glucose release from olive stones from 0.015 to 0.056 kg/kg after 144 h of hydrolysis [63].

The response surfaces in Fig. 6B have enzyme load fixed at a high level (22 FPU/g); lowering enzyme load will give surfaces with identical shapes but lower predicted responses. Enzyme load had a significant positive linear effect, with a large estimated increase in predicted glucose hydrolysate of 5.49 g/L when enzyme load is increased from 11 to 22 FPU/g (0.499 × (22-11) = 5.49 g/L, Table 5). Expressing this fixed increase as a percentage increase in yield for MA and FA separately, for fixed temperature 180 °C and concentration 0.6 mol/L, the predicted increase in glucose yield is 22.2 % for MA (from 24.8 g/L to 30.3 g/L) and 38.7 % for FA (from 14.2 to 19.7) when enzyme load is increased from 11 to 22 FPU/g (based on Table 5). Repeating these calculations for ethanol, the large estimated increase in predicted ethanol is 2.73 g/L when the enzyme load is increased from 11 to 22 FPU/g (0.248 × (22-11) = 2.73 g/L, Table 5). Expressing this fixed increase as a percentage increase in yield for MA and FA separately, for fixed temperature 180 °C and concentration 0.6 mol/L, the predicted increase in ethanol is 26.4 % for MA (from 10.7 g/L to 13.4 g/L) and 42.6 % for FA (from 6.1 g/L to 8.7 g/L) when the enzyme load is increased from 11 to 22 FPU/g.

Tween 80 is a common additive used in cellulose saccharification, known for enhancing the saccharification of various lignocellulosic biomasses. It achieves this by reducing protein aggregation and minimizing the inhibitory effects of lignin, thereby improving enzyme accessibility to cellulose [64–66]and enhancing surface adsorption [67]. In the case of microwave-assisted organic acid (OA) treatments, Tween 80 showed no significant effects on glucose hydrolysate or ethanol production, nor did it interact with any other factors at the levels tested in this experiment (ranging from 0 to 0.09 g/g). This lack of effect may be attributed to the specific conditions of the biomass, such as porosity and hydrophobicity, as surfactants are known to influence enzyme-substrate binding under certain circumstances. It is important to note that pretreatment methods that increase biomass porosity and reduce lignin content may facilitate a more effective enhancement of saccharification by Tween 80 [68]. Therefore, further investigation is needed to understand how our pretreatment impacts the characterization of the biomass, as well as to explore the effects of higher concentrations of Tween 80.

MA resulted in more glucose hydrolysate than FA, and the interaction between pretreatment temperature and OA type is significant, with glucose hydrolysate strongly dependent on temperature in the presence of MA (Fig. 6B; the predicted level of glucose hydrolysate is 15.6 g/L at 120 °C and 30.3 g/L, 85 % of the cellulose fraction after pretreatment, at 180 °C), and less so with FA (for OA concentration at 0.6 mol/L, enzyme load at 22 FPU/g, Table 5 and front of left surface plot in Fig. 6B; only reaching 19.7 g/L, 42.0 % of the cellulose fraction after pretreatment). Repeating these calculations for ethanol, MA resulted in more ethanol compared to FA and the interaction between pretreatment temperature and OA type is significant, with ethanol strongly dependent on temperature in the presence of MA (Fig. 6 B; the predicted level of ethanol is 6.7 g/L at 120 °C and 13.4 g/L, 73.7 % of the maximum theoretical yield, based on glucose content after pretreatment, at 180 °C), and less so with FA (for OA concentration at 0.6 mol/L, enzyme load at 22 FPU/g, Table 5 and front of right surface plot in Fig. 6B; only reaching 8.9 g/L, 37.8 % of the maximum theoretical yield, based on glucose content after pretreatment). On the other hand, OA concentration had a small positive effect on glucose hydrolysate and ethanol yield and exhibited a minor interaction with pretreatment temperature. Furthermore, the effect of pretreatment temperature showed a slight increase when the concentration of MA or FA is higher, as evidenced by comparison of the back and front surfaces in Fig. 6B.

Glucose hydrolysate and ethanol production showed the same dependence on pretreatment factors across all levels of enzyme load and Tween 80 during enzymatic hydrolysis. A supplementary investigation illustrates the positive linear dependence of responses from the later stages of the process on responses from the pretreatment stage when considered separately for each level of enzyme load for runs with OA (shown for glucose hydrolysate in Fig. 6C). This dependence was expected, since the fitted models for the different stages have the same dominant effects, temperature, and OA type, and these effects have identical direction, with high levels of responses obtained with MA at high temperature. The results shown in Fig. 6C can be attributed to hemicellulose degradation and breaking of lignin-cellulose bonds during the pretreatment process. This makes the cellulose more accessible for enzymes during the saccharification process. This could lead to an increase in the amount of sugar available for the fermentation process. It might be that the combination/separate pretreatment hydrolysate and the saccharification hydrolysate could result in less sugar loss for the fermentation process. In our study, we did not consider fermenting the sugar released after the pretreatment stage. The liquid fraction could feasibly be fermented if detoxified and/or pentose-fermenting strains (e.g., *Pichia stipitis, Zymomonas mobilis*) are employed. Therefore, we

Table 5
Fitted models display the effects of pretreatment parameters on glucose released after enzymatic saccharification and on ethanol production of OMSW treated with maleic acid and formic acid.

OA type	Fitted models
Maleic acid	Glucose hydrolysate (g/L) = – 22.4280 + 0.2210 × Temperature + 2.0125 × OA conc. + 0.4990 × Enzyme load +0.1261 × (Temperature – 146.2500) × (OA conc. – 0.4000) Ethanol (g/L) = – 10.2660+ 0.0956 × Temperature + 0.9100 × OA conc + 0.2480 × Enzyme load + 0.0670 × (Temperature – 146.2500) × (OA conc – 0.4000)
Formic acid	Glucose hydrolysate (g/L) = – 0.8400 + 0.0415 × Temperature + 2.0126 × OA conc. + 0.4990 × Enzyme load +0.1261 × (Temperature – 146.2500) × (OA conc – 0.4000) Ethanol (g/L) = – 0.4700 + 0.0160 × Temperature + 0.9100 × OA conc + 0.2480 × Enzyme load + 0.0670 × (Temperature – 146.2500) × (OA conc – 0.4000)

propose to investigate that aspect in near-future research direction. The yeast *Saccharomyces cerevisiae* is the most widely used microorganism for bioethanol production due to its high ethanol yield and tolerance to stress. However, it cannot naturally ferment pentose sugars, such as xylose and L-arabinose, which are abundant in lignocellulosic biomass [69]. To address this limitation, efforts over several decades have focused on either identifying or developing yeasts capable of fermenting both hexoses and pentoses under industrial conditions. This has taken two main approaches: discovering naturally occurring pentose-fermenting microorganisms from lignocellulose-rich environments, and genetically engineering these native microorganisms or *S. cerevisiae* to enhance pentose utilization. Each strategy has its advantages and disadvantages, and several recent reviews detail the progress in genetic engineering to improve ethanol yield and strain robustness [70].

Our overall conclusion, based on the absence of interactions between the four factors of the pretreatment process and the two factors of the saccharification processes investigated in this study, emphasizes that the two processes can be optimized separately in future research. As commented earlier, one of the significant aspects of this research is that it offers a conceptual approach to independently optimizing the pretreatment and enzymatic hydrolysis stages. The same six-factor Design of Experiments (DoE) model and response surface analyses can be used to simplify process optimization and reduce experimental workload in other lignocellulosic feedstocks and pre-fermentation.

3.5. Trade-off and joint optimization of all stages

The results for loss of sugars, formation of inhibitors, toxicity to yeast, cellulose saccharification, and ethanol production showed that

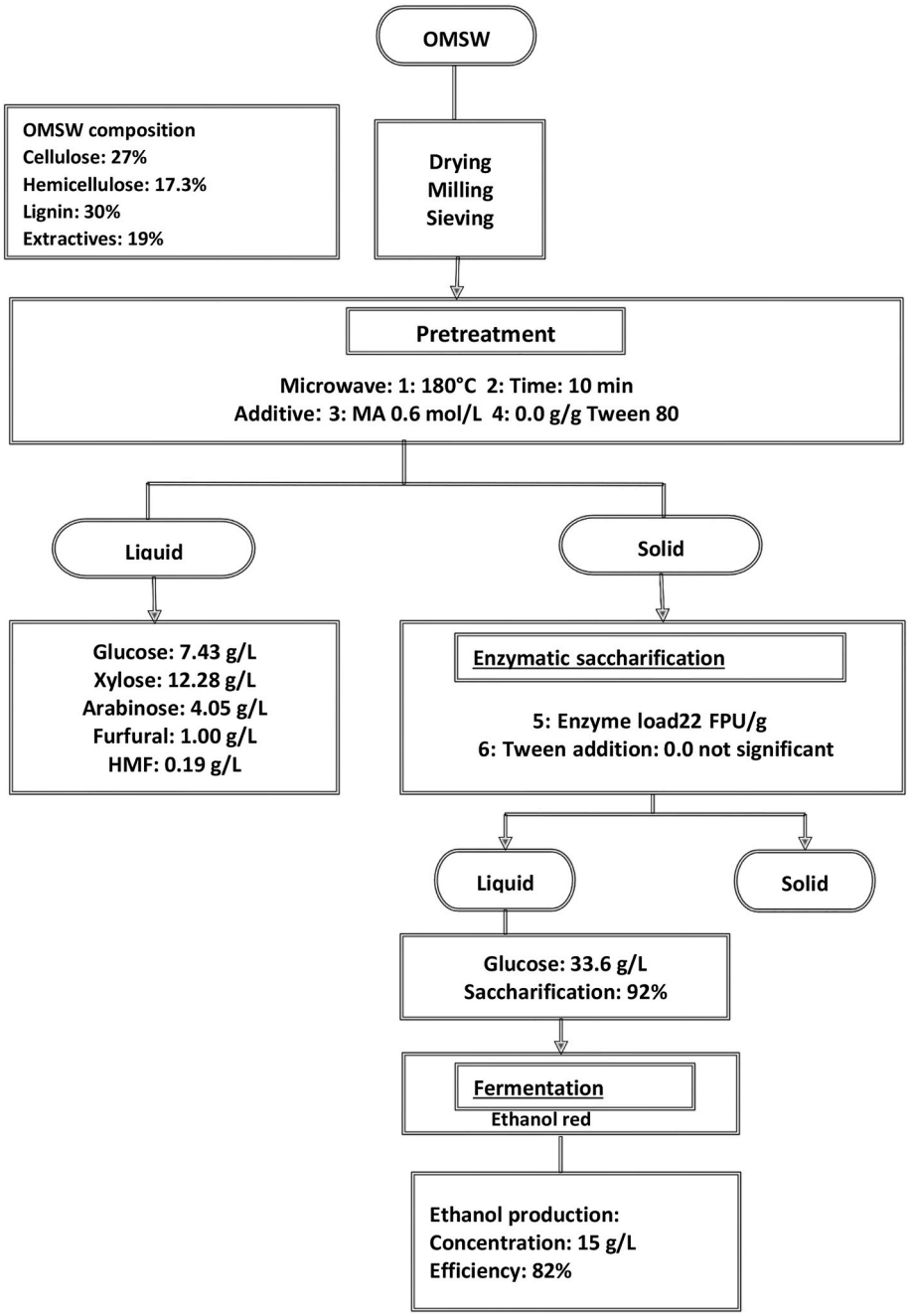


Fig. 7. Diagram for the general OMSW component and cellulose fraction under optimum conditions.

high-temperature pretreatment with MA gave the best ethanol yield, but also resulted in high loss of sugars.

These results align with the strength and dicarboxylic properties of MA, known to provide selectivity toward the hydrolysis of hemicelluloses [46,71,72]. Potentially, this tradeoff could be overcome by avoiding the separation of pretreatment solids and liquids, thus enabling the utilization of sugars lost to the latter by yeasts (17) or bacteria, i.e., Escherichia coli, ethanologenic recombinant microorganisms, which can consume both hexose and pentose sugars with high tolerance to inhibitory compounds, after the detoxification stage [58]. The results of yeast inhibition, furfural, and HMF found in the pretreatment liquid for runs with MA suggest that this concept is valid, as no inhibition of yeast growth was observed, and lower than inhibitory furfural and HMF concentrations were found. The meaningful sugar release to the pre-treatment liquid (~18 % of the releasable glucose and ~94 % of the pentose sugars; Fig. 7) and the observed low toxicity of MA suggest that, in principle, these sugars could be fermented by microorganisms capable of utilizing pentoses [69], although such are still a rarity in the bioethanol industry (hence the use of Ethanol Red® here) [70]. This subject will be investigated in further studies. Of particular interest are the interactions between the pretreatment and saccharification factors, which were included in initial models but found insignificant. The absence of such interactions indicates that the pretreatment and saccharification stages can be optimized separately without completing the whole process, making optimization much simpler. In addition, recycling the MA (possibly using low-temperature crystallization; [73]) could enhance the economic feasibility of the process [55,72,74] [55,72,74]. Additional results that can lead to cost reduction are that at the recommended levels of the other four factors, prolonged MW pretreatment is unnecessary, allowing for lower energy consumption, and Tween 80 is unnecessary (Table 6).

3.6. Comparison of observed responses with those observed in similar experiments

The observed percent of conversion of cellulose to glucose obtained after enzymatic saccharification ranged from 20 to 94 % for the MA-treated OMSW and 17–42 % for the FA-treated OMSW, slightly higher than previously reported for OMSW with FA and 10 min MW pretreatment ([1]; although OMSW composition was different). The saccharification rate was also higher than the 52–78 % obtained from Water Hyacinth with much longer pretreatment (hours; [59]), or 82 % for wheat straw after 50 min, 170 °C, 0.089 mol/L maleic acid [55]. Another study used sawdust of deodar cedar to obtain 29.20 g/L reducing sugar after thermochemical pretreatment and sequence enzymatic hydrolysis, using optimized conditions: 1.5 % HCl, 10 % biomass loading, and 30 min, followed by enzymatic hydrolysis with cellulase and xylanase [75]. The fermentation process for the hydrolysate was performed with *Saccharomyces cerevisiae* and *Pichia stipitis* to obtain 14.25 g/L ethanol with 95.68 % conversion efficiency. The saccharification rate of 84 % was obtained from rice straw with 8 % sodium perborate pretreatment (80 °C, 4 h), followed by enzymatic hydrolysis [76]. The maximum ethanol concentration was 15.29 g/L, with 91.96 % fermentation efficiency [76]. Table 7.

A wide range of levels of ethanol was observed, with the best runs showing 13–15 g/L ethanol (64–88 % of the maximum theoretical yield, based on glucose content after pretreatment), equivalent to 8.4–9.3 g ethanol per 100 g raw OMSW, or conversion of 55–61 % of the cellulosic fraction of the OMSW. These results are similar to previously published results (9.11 g ethanol per 100g OMSW; [1]) and can probably be further enhanced by improving fermentation conditions (in Ref. [1], 87.7 % here, 88.3 % of the theoretical maximum).

4. Conclusion

Combined microwave and organic acid pretreatment offer several advantages and increased process efficiency over conventional heating. In this study, we investigated the effect of four pretreatment factors (temperature, time, and OA type and concentrations) and two saccharification factors (enzyme load and Tween 80 addition) and interactions between all the six factors on loss of sugars, formation of inhibitors, loss of biomass toxicity to yeast, cellulose saccharification, and ethanol production. Within the investigated ranges, the analysis showed that MW temperature, OA type, and enzyme load had the most significant effects on ethanol production, with considerable interaction between OA type and MW temperature. Enzyme load had no interaction with any of the pretreatment factors. Maximum predicted ethanol of 13.4 g/L was achieved with the temperature at a high level (180 °C), MA at a high concentration (0.6 mol/L), and enzyme load at a high level (22 FPU/g treated OMSW). Furthermore, as no inhibition of yeast by pretreatment was evident with MA, it is suggested that by using this specific acid, the separation of the pre-treatment liquids from the solids might be avoided, lowering complexity and utilizing the "lost" sugars, although the effect of the pre-treatment liquid on the enzymatic hydrolysis should be studied. Key findings relevant to planning future experimentation, including such with other biomass/pretreatment, were the absence of significant interactions between factors from the two stages of the process, indicating that they could be optimized separately; the absence of significant quadratic effects (in all but one model); the very high correlation observed between glucose hydrolysate and ethanol, indicating that the saccharification process could be optimized without completing fermentation; the poor results for the control runs without OA.

CRediT authorship contribution statement

Hiba N. Abu Tayeh: Writing – original draft, Project administration, Investigation, Formal analysis, Conceptualization. **Jacqueline A. Asscher:** Writing – review & editing, Methodology, Formal analysis. **Hassan Azaizeh:** Writing – review & editing, Supervision, Conceptualization. **Yoram Gerchman:** Writing – review & editing, Supervision,

Table 6
Factors levels in the experiment and the best and alternative levels obtained during the analysis.

Factor	Units	Levels (coded values for continuous factors)			Best levels (giving maximum predicted ethanol)	Alternative levels (giving high predicted ethanol at lower cost)
		−1	0	+1		
Temperature	°C	120	150	180	180	180
Time	minutes	4	7	10	Not significant	Not significant
OA type		None, formic, maleic			Maleic	Maleic
OA concentration	mol/L	0.2 (for OA type = Formic or Maleic)	0.4	0.6	0.6 (negligible effect)	0.4
Enzyme load	FPU/g treated OMSW	11	16.5	22	22	16.5
Tween 80	g/g treated OMSW	0	0.045	0.09	Not significant	Not significant

OA: Organic acid.

Table 7
Comparison of the observed responses with those observed in similar experiments.

Reference	Biomass	Pretreatment	Cellulose content (%)	Enzyme load	Max saccharification (%)	Max glucose (g/L)	Yeast	Max EtOH (g/L) – Yield (g ethanol/g cellulose)	EtOH % of theoretical (Glucose based)
[63]	Olive stones	Liquid hot water, 225 °C, 0min	25.30	8280 IU cellulase, 345 IU β-glucosidase	82.80	N/A	*Pachysolen tannophilus* ATCC 32691	2.50 – N/A	N/A
[18]	Olive stones	High temperature, diluted acid	29.90	40 (FPU/g)	80.00	Diluted with water up to a glucose concentration of 20.60 g/L	*Pachysolen tannophilus* ATCC 32691	10.80 – 0.523	~ 100
[1]	OMSW 3 phases	Microwave 0.6 mol/L formic acid, 140 °C, 10 min	N/A	15 (FPU/g)	90.00	30.40	Ethanol Red	15.9 – 0.46	87.70
[75]	Deodar sawdust	1.5 % HCl, 30 min, 121 °C, 1 bar	33.90	29 U/g Cellulase 9 U/g Xylanase		29.20	*Saccharomyces cerevisiae* (MTCC-36) and *Pichia stipitis* (NCIM-3498)	14.25 – 0.1425	95.68
[76]	Rice straw	8 % sodium perborate, 80 °C, 4 h	34.40	20 FPU/g	84.00	702.40 g/kg	*Saccharomyces cerevisiae* SHY07-1	15.29 (SScF) – 1.10	91.96
[77]	Olive Tree Pruning	1 % of Ca $(OH)_2$ 5 % (w/v), 120 min, 160 °C	31.88	cellulase (ATCC 26921) (≥700 units/g) and hemicellulose (0.3–3 units/mg)		10.50	*Saccharomyces cerevisiae BSc*114	1.42 – 0.088	13.50
[78]	OMSW (2-phase)	Fungi *Fusarium oxysporum*	N/A	2.88 IU/mL cellulase 0.57 IU/mL laccase 8.43 IU/mL peroxidase		N/A		2.47 – 0.84	CBP
Present work	OMSW 3 phases	Microwave 0.6 mol/L maleic acid, 180 °C	27.00	22 (FPU/g)	92.40	33.60	Ethanol Red	15.00 – 0.34	88.30

N/A – not available, SScF - simultaneous saccharification and co-fermentation, CBP - Consolidated Bioprocessing.

Conceptualization.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biombioe.2025.108725.

Data availability

Data will be made available on request.

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